

physics modelling agent

progress report

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supervisor 2: Prof. Dr. Samir Lounis

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outline

- 1 goal of the project
- 2 key definitions
- 3 architecture
 - constraints
 - crewai
- 4 design process
 - general architecture
 - challenges during the design process
- 5 example workflow: The Edelstein Effect
- 6 next steps
- 7 summary

aim of the project

what the “physics modelling agent” should be

A configuration of “intelligent systems” which assists a human researcher in building a model to answer a research question.

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- “physics modelling agent” takes the input:
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reports findings:

- human readable **model description**
- executable **numerical implementation**

definitions of LLM and agentic AI

what is an LLM

- “Transformer language models” [19] with at least billions of parameters (no clear cut-off) [19]
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what is an agentic AI

- an AI with enhanced capabilities including: strategic planning, (external) tools usage, collaboration with other agents [3]

architectural foundations

constraints

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- access via remote servers with valid API key [9]

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design choices

- use agentic AI to harness strategic planning capabilities and tool usage
- Scalable AI Accelerator (**SAIA**) from the GWDG[10]
- access via remote servers with valid API key [9]
 - use an existing framework for AI orchestration
 - save time on the software engineering part of the project
 - focus on definition of AI agents and configuration of their behaviour



Figure: framework chosen: crewai [6]

Agents[4]

accesses an agentic AI model to perform a task

Agents⁴

accesses an agentic AI model to perform a task

- allows to access any AI-models we have access to (SAIA)
- behaviour can be defined via parameters (role, goal, backstory, creativity)
- tools can be added to enhance capabilities (e.g. web search, code execution)

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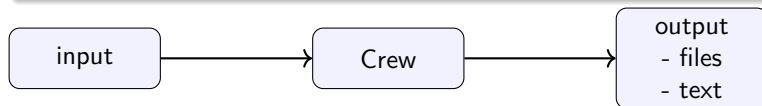
- defined by task description and expected output
- is assigned an agent (execution merges agent and task description)

the Crew[5]

contains all agents with their tasks and handles the interactions between agents

the Crew⁵

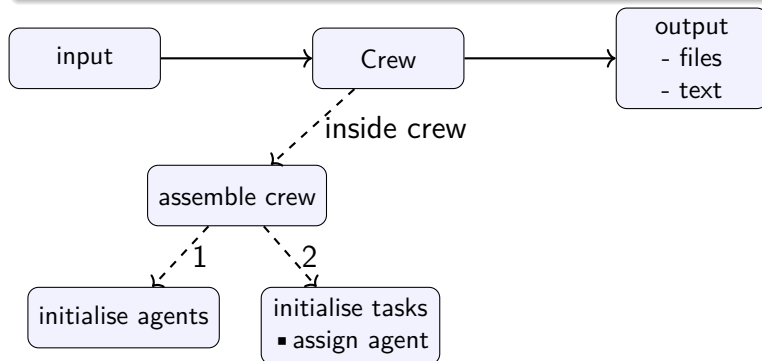
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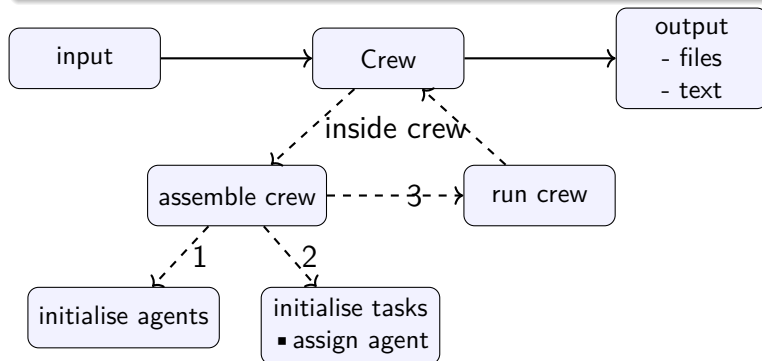
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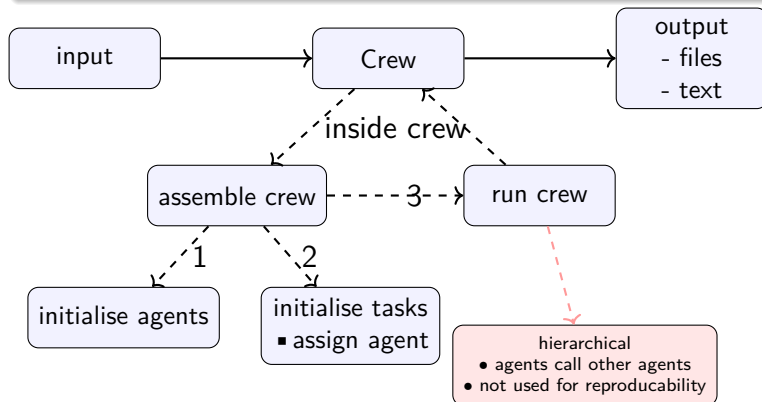
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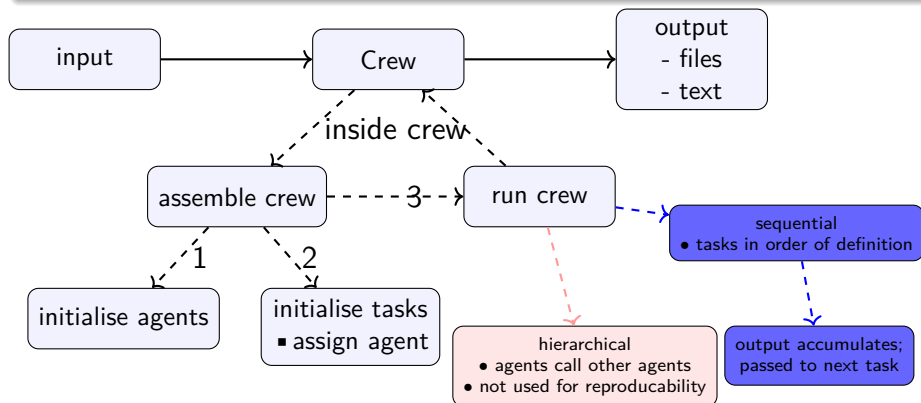
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orchestrating agentic AI's

free form modelling request

Figure: general vision for the steps the crew should perform

orchestrating agentic AI's

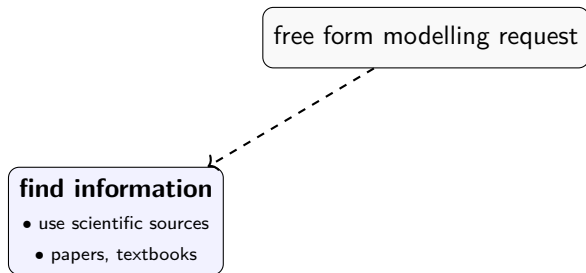


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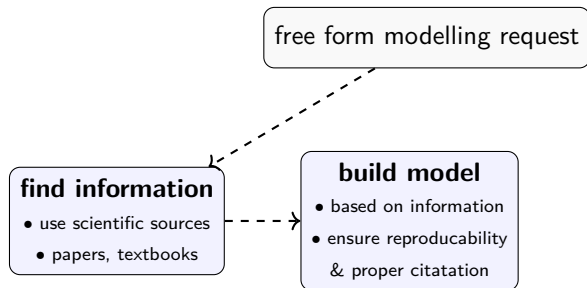


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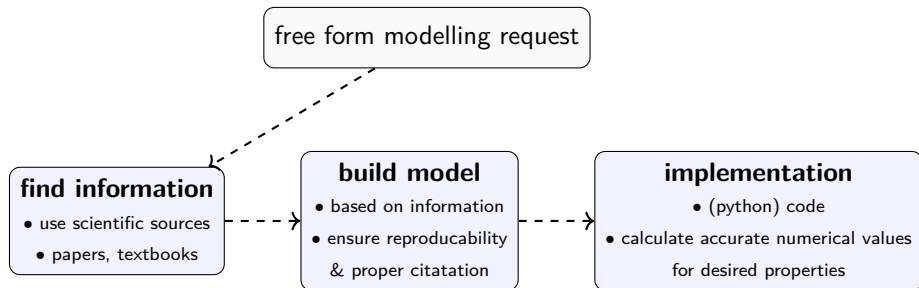


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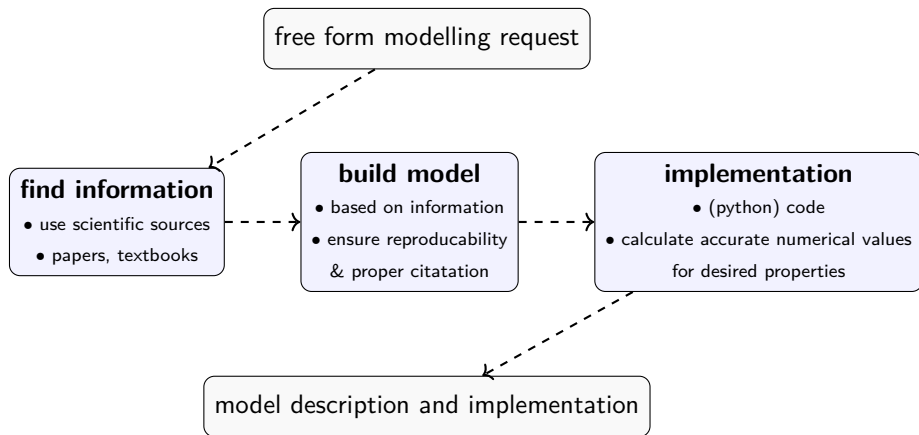
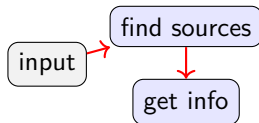


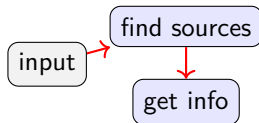
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information finding



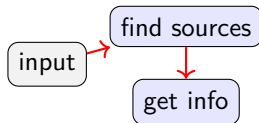
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problems encountered



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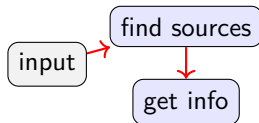
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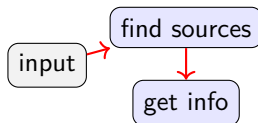
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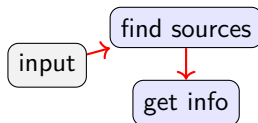
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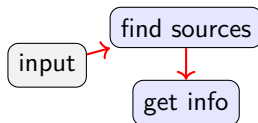
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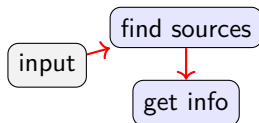
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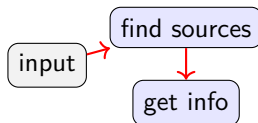
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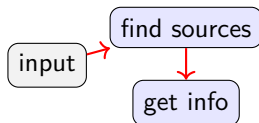
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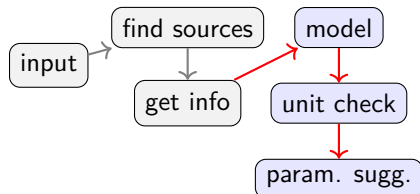
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 - ✓ separate the task performing the arXiv request from the crew
 - ⇒ no new requests when testing other tasks

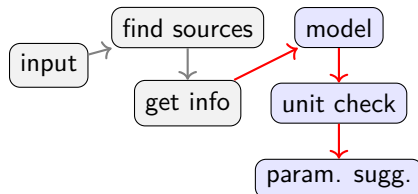


modelling



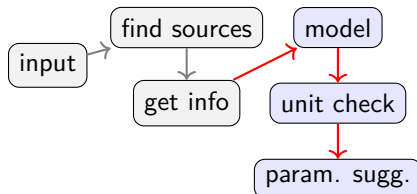
the unit problem

- 1 task outputs formula from the sources; theoretical formulas often use different definitions of parameters



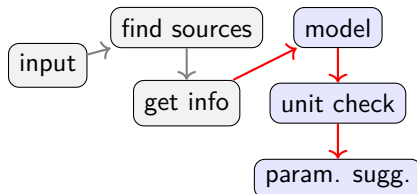
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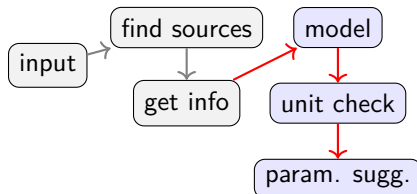
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starting parameters

- for realistic results realistic start parameters are necessary



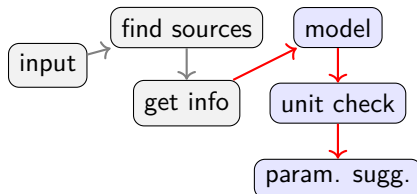
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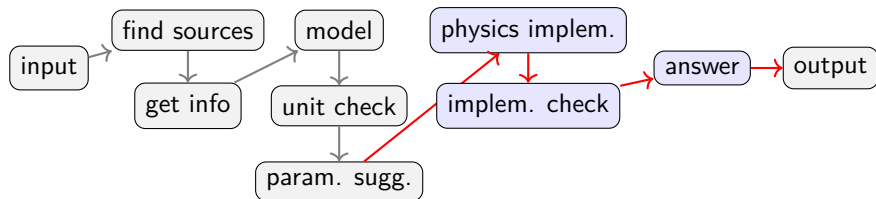
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- ⇒ agent for suggesting reasonable starting parameters



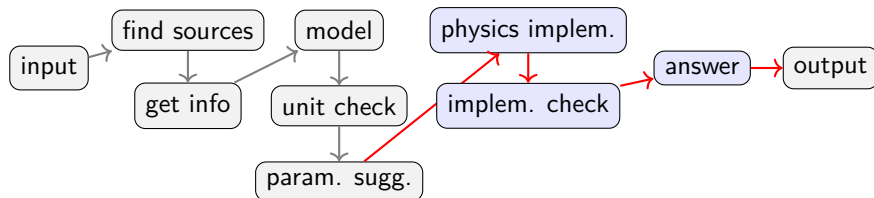
numerical implementation



numerical implementation

minor challenges

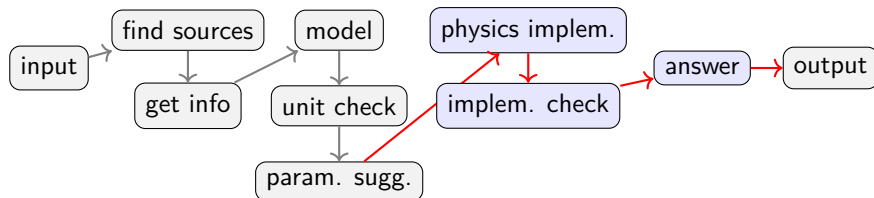
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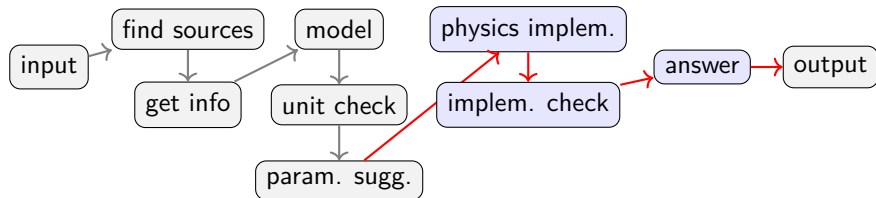
numerical implementation

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answer: provide a short result at runtime

- needed for automatic evaluation of the system (benchmarking)



example: input/modelling request

input

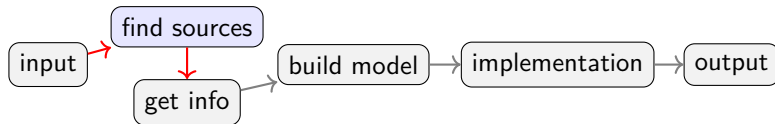
aim : “Calculate the Edelstein effect for a Rashba fermion (at the Gamma point of the Brillouin zone).

Compute the magnetization magnitude and direction of different directions and magnitudes of the applied electric field.

Consider how the result depends on relevant parameters of the model (e.g. chirality, fermi velocity, spin-orbit coupling strength) and make explicit graphics.”

topic : Edelstein-Effect

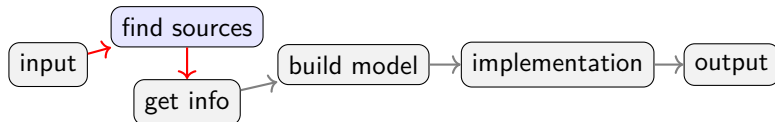
papers found



papers found

agent configuration:

- no creativity (temperature = 0)
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- tools: arXiv_Papers_With_Backoff



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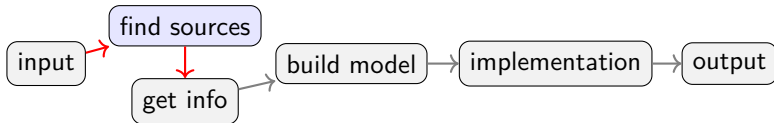
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papers found ([view full report](#))

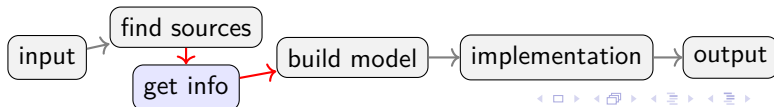
“Edelstein Effect in Isotropic and Anisotropic Rashba Models”[12]

“Spin and orbital Edelstein effect in spin-orbit coupled
noncentrosymmetric superconductors”[1]

“Spin and orbital Edelstein effect in a bilayer system with Rashba
interaction”[14]



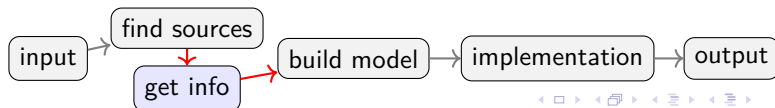
information extraction



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extracted information ([view full report](#))

$$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\vec{p} \times \vec{\sigma})$$

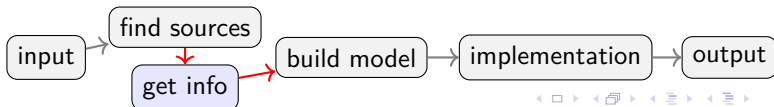
Rashba-Hamiltonian

$$M_y = \frac{\mu_b |e| \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$$

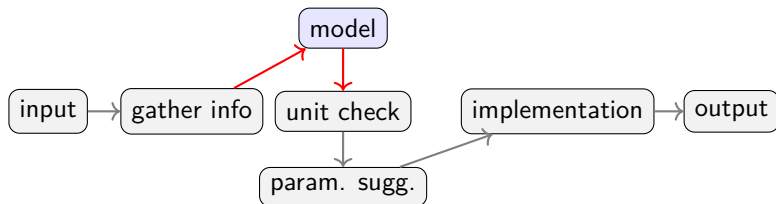
magnetization in HDR

$$M_y = \frac{\mu_b |e| \tau}{2\pi} \sqrt{m^2 \alpha^2 + 2m E_F} [\hat{z} \times \mathbf{E}]_y$$

magnetization in LDR



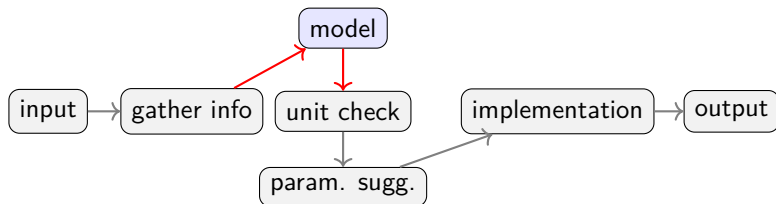
build model



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agent configuration:

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- LLM model: deepseek-r1-distill-llama-70b[13]; (optimized for logic & reasoning)
- tools: this LLM is unable to use tools



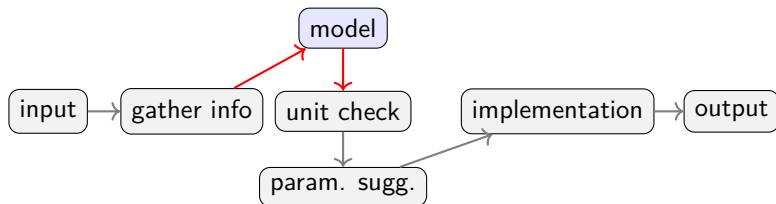
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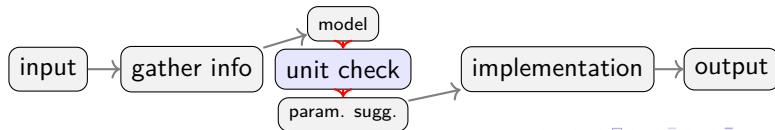
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extracted information ([view full report](#))

restates results from information extraction

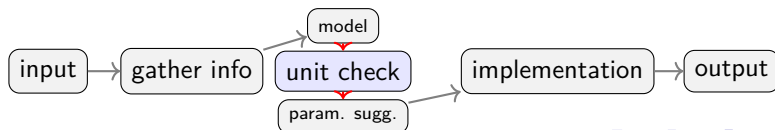


unit check



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unit check

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unit corrections ([view full report](#))

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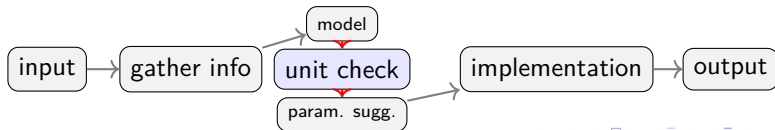
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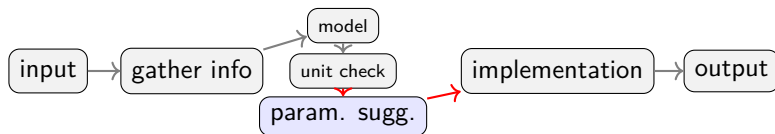
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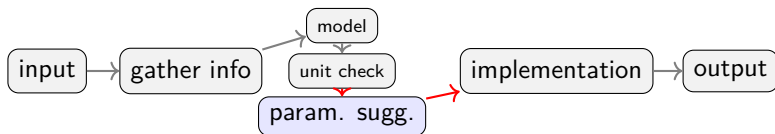


parameter suggestion



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suggested starting parameters ([view full report](#))

$$m \approx 6,10 \times 10^{-32} \text{kg}$$

effective mass

$$\alpha = 5,0 \times 10^4 \text{m/s}$$

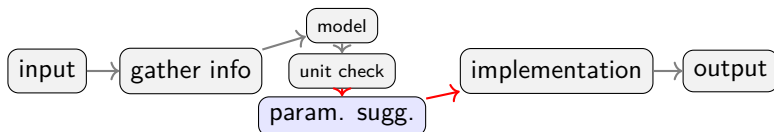
Rashba Coupling Strength

$$\tau = 5,0 \times 10^{-13} \text{s}$$

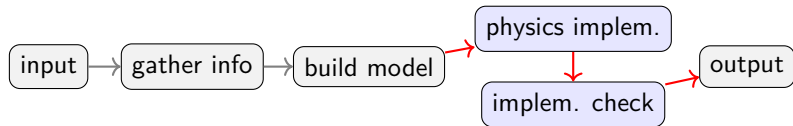
transport time

$$E_x = 1,0 \times 10^5 \text{V/m}$$

applied electric field



numerical implementation I



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results for magnetisation ([view full report](#))

Isotropic HDR Magnetization:

$$M_y = 3,25 \times 10^{-9} \text{ A/m}$$

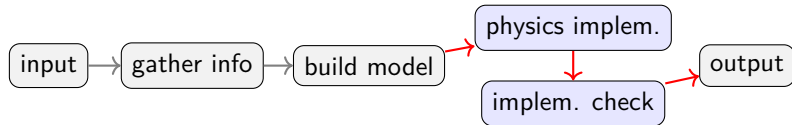
Isotropic LDR Magnetization:

$$M_y = 4,65 \times 10^{-35} \text{ A/m}$$

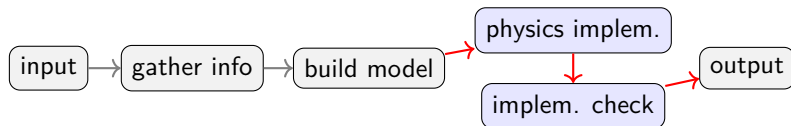
Anisotropic Magnetization Components:

$$M_x = 2,04 \times 10^{-8} \text{ A/m}$$

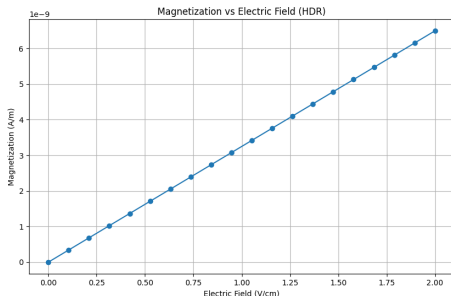
$$M_y = 2,04 \times 10^{-8} \text{ A/m}$$



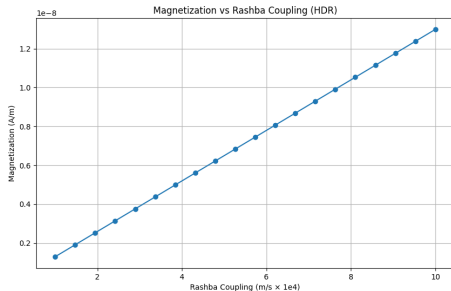
numerical implementation II



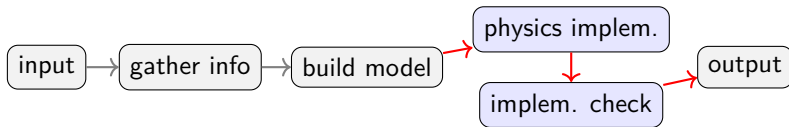
numerical implementation II



(a) magnetisation over electric field



(b) magnetisation over Rashba param. α



next steps

benchmark the “physics-modelling-agent”

a method to measure the systems performance

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- ⑤ sources: how does the number (and possibly type) of sources influence the systems results?

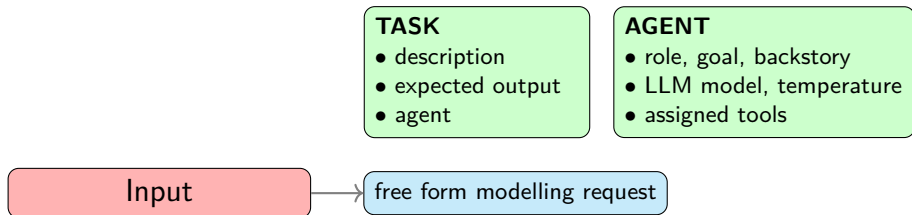
summary

Input

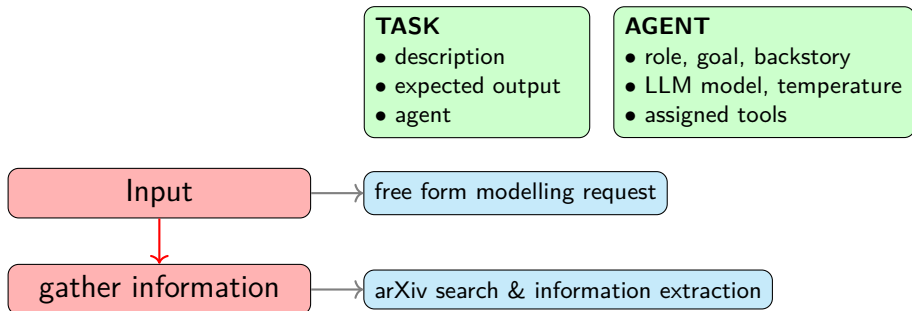


free form modelling request

summary



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summary

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- expected output
- agent

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- role, goal, backstory
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dimensional analysis

implement model

executable python code calculations

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